

HARLEEN BUTTAR

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EDUCATION

M.S. Applied Data Science, University of Chicago | GPA: 3.83

August 2026

Coursework: Machine Learning, Big Data & Cloud Computing, Time Series Analysis, Algorithmic Marketing, Statistical Models, Data Engineering, NLP

B.S. Data Science, University of Chicago

June 2025

DATA SCIENCE EXPERIENCE

Data Science Capstone Consultant

March 2026 - August 2026

Rakuten Advertising, University of Chicago

- Collaborated as an equal member of a four-person team benchmarking five time series foundation models (Chronos, TimesFM, TTM, MOMENT, Moirai 2.0) against parameter-efficient fine-tuning techniques (LoRA, QLoRA, DoRA, BitFit) for retail demand forecasting on NielsenIQ receipt-level data across Luxury Brand Retail, Luxury Department Retail, and Travel verticals; implemented the TTM and Moirai 2.0 classes end to end
- Built a full-stack forecasting dashboard (React/Vite/Recharts, FastAPI) supporting multi-model and fine-tuning-method comparison, overlay charts with p10/p90 uncertainty bands, CSV export, and persistent view state, deployed to Vercel and Render
- Resolved production deployment issues spanning Python version pinning, CPU-only PyTorch, dependency conflicts, and a cross-platform filename casing bug invisible on macOS that hard-failed the Linux build; coauthored the academic paper, revising after reviewer feedback to foreground substantive tensions across approaches

Data Science Project Manager

June 2025 - September 2025

CharacTour, Chicago, IL

- Directed 3 data science interns analyzing 10M+ consumer records; built Python NLP pipelines from TikTok transcripts detecting cluster-specific speech patterns to power identity-driven AI recommendations for 23 personality segments
- Developed scalable automated diarization pipeline to slice, transcribe, and assign speaker identities across long-form video scenes, reducing manual tagging by 80%

Team Member, Data Science Clinic

January 2025 - June 2025

Internet Equity Initiative, University of Chicago, Chicago, IL

- Built and deployed interactive broadband equity dashboard in Plotly and Plotly Dash to visualize disparities in internet access across geographies and demographics
- Partnered with policy researchers to translate continuous data from thousands of devices into actionable insights informing broadband infrastructure and digital equity recommendations

GRADUATE PROJECTS

QLoRA Recipe Chef (Machine Learning 2)

March - June 2026

- Developed an end-to-end recipe generation system by fine-tuning the Qwen2.5-1.5B-Instruct model with QLoRA, 4-bit quantization, and LoRA adapters on the RecipeNLG dataset, enabling controllable recipe generation conditioned on cuisine and dietary preferences while training only 0.6% of model parameters
- Built a complete ML evaluation and deployment pipeline, including custom cuisine/diet labeling, automated metrics (ROUGE, BERTScore, perplexity, ingredient fidelity, knowledge retention), and an interactive Gradio app; optimized Drive-backed caching to cut experiment rerun time from ~90 to ~5 minutes

Clothing Fit Prediction & Size Recommendation (RentTheRunway / ModCloth)

January - March 2026

- Built 3-phase ML pipeline across 192,000+ transactions: unsupervised segmentation (K-Means, GMM, DBSCAN, K-Modes, PCA, LDA), 7-classifier fit prediction (SVM, Random Forest, CART, KNN, Naive Bayes) with SMOTE, and collaborative filtering recommender
- Benchmarked all classifiers on multiclass fit prediction (Small/Fit/Large) with per-class precision, recall, and F1; tested cross-retailer generalization by training on ModCloth and evaluating on RentTheRunway

UK Fruit and Vegetable Price Forecasting (Time Series Analysis)

January - March 2026

- Engineered 184-feature weekly dataset incorporating agricultural indices, UK fuel prices, and freight indices with energy crisis shock indicators; benchmarked ARIMA, Holt-Winters, VAR, LM-ARIMA, SARIMA, ARIMAX, DeepAR, and Mamba across 20 commodities with Granger causality and stationarity testing

Large-Scale Sentiment-Rating Mismatch Detection (Big Data)

September - December 2025

- Built distributed PySpark NLP pipeline on Google Cloud Dataproc applying TF-IDF, Word2Vec, K-Means, LSH, and LDA at scale on Amazon review data; managed schema validation, data type compatibility, and fault tolerance with outputs written to GCS

TEACHING EXPERIENCE

Teaching Assistant, Data Engineering & Data Science Clinic — University of Chicago

September 2025 - June 2026

- Support graduate students through pipeline design, SQL, Python, cloud infrastructure, and GitHub; guide code quality standards, collaborative development practices, and technical communication across demanding coursework

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, scikit-learn, PySpark, Plotly, Selenium, NLTK, Gensim, VADER), R, SQL (Postgres, MySQL, SQL Server), DuckDB, Java

ML & Analytics: Supervised/Unsupervised ML, NLP, Causal Inference, Time Series Forecasting, Recommendation Systems, Feature Engineering, Statistical Modeling

Tools & Platforms: Tableau, Plotly Dash, GitHub, GCP, AWS, Databricks **Languages:** English, Punjabi, Spanish (all fluent)